Signorini's Problem

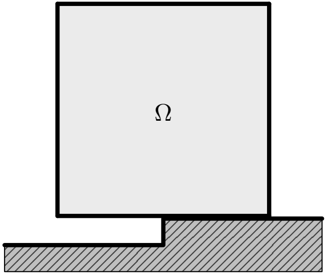
Max Stuthmann

March 13, 2026

Content

Rough Problem Setup

- *Body:* $\Omega \subset \mathbb{R}^d$, $d \in \{2,3\}$ open, bounded, connected with Lipschitz boundary $\Gamma := \partial\Omega$.



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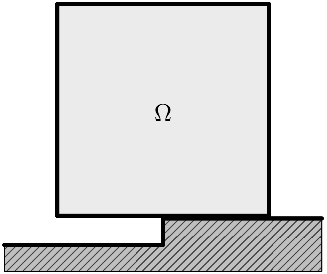

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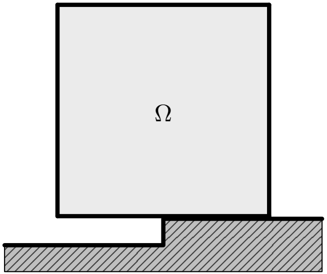

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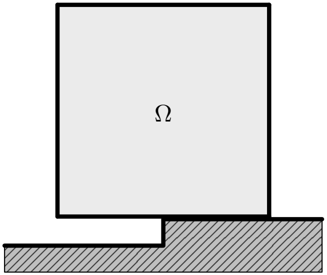

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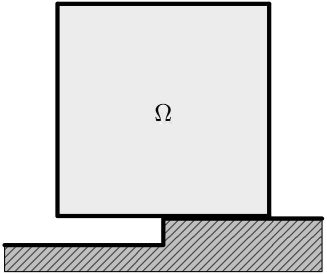
- Ω may be subject to elasticity properties and internal/external forces.
- **Not in equilibrium configuration**.



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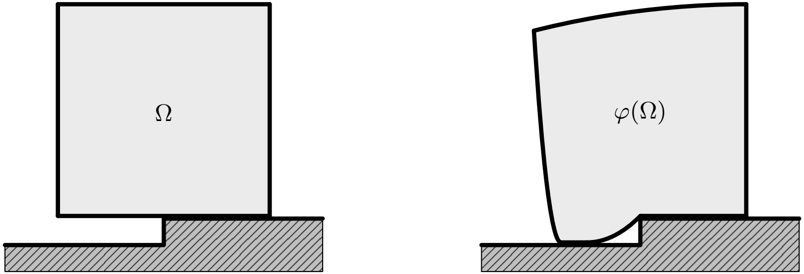

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- Let $\varphi(\Omega)$ denote Ω in equilibrium configuration.
- $\varphi : \overline{\Omega} \to \mathbb{R}^d$ maps $\overline{\Omega}$ onto $\varphi(\overline{\Omega})$ and can be written as

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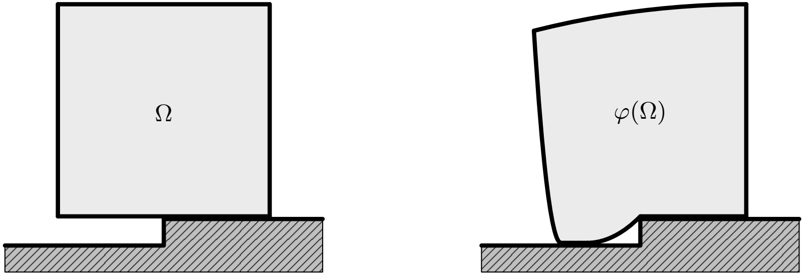

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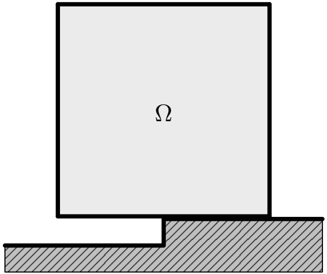
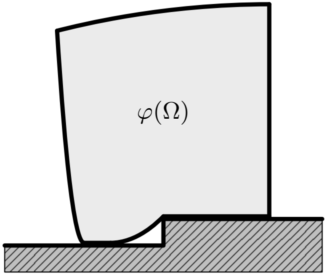
- **We aim to find the *displacement function* u.**

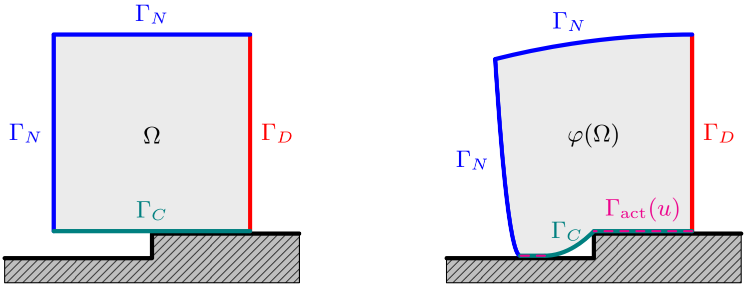




Content

Boundary Decomposition and Preview on the Boundary Conditions

- Decompose $\Gamma = \partial\Omega$ into three relatively open and pairwise disjoint parts $\Gamma = \overline{\Gamma_D} \,\dot\cup\, \overline{\Gamma_N} \,\dot\cup\, \overline{\Gamma_C}$ with $|\Gamma_D|, |\Gamma_C| > 0$.



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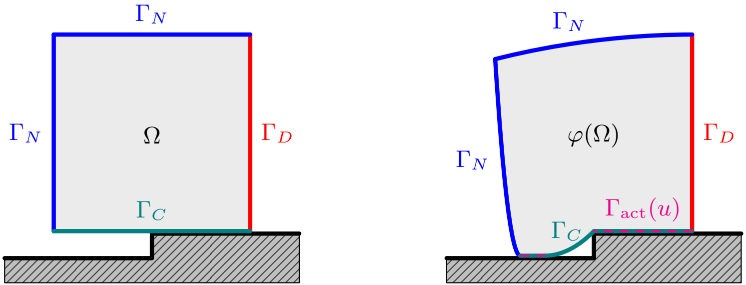

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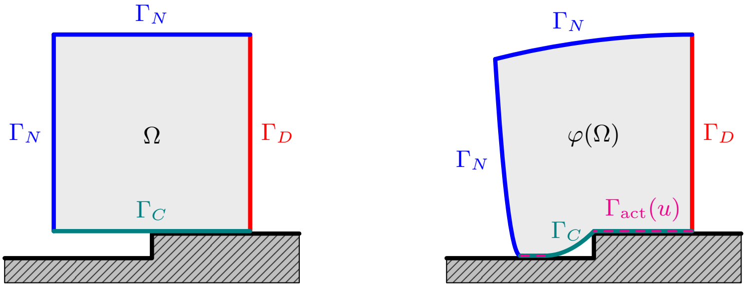

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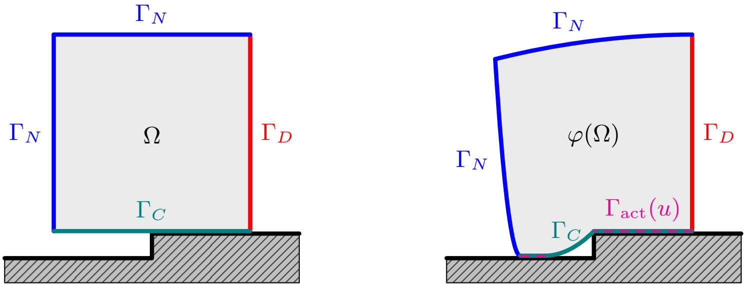

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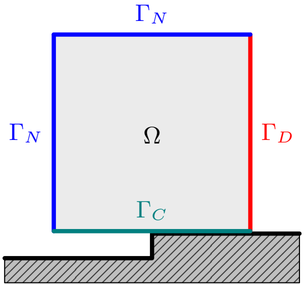

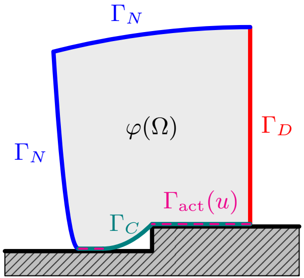

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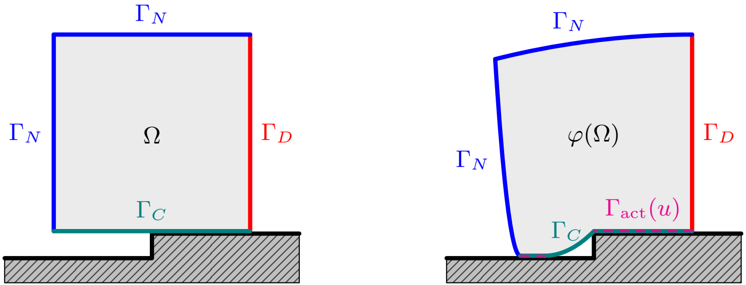
- Rigid foundation can be describe by gap-function $g : \Gamma_C \to \mathbb{R}$. We assume $g = 0$.



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- If $\mathcal{A} : \Omega \to \mathcal{L}(\mathbb{S}^d; \mathbb{S}^d)$ is a fourth order tensor and $s \in \mathbb{S}^d$, we use the *double contraction*

$$(\mathcal{A}(x) : s)_{ij} := \sum_{k=1}^{d} \sum_{l=1}^{d} \mathcal{A}_{ijkl}(x) s_{kl}, \qquad s : \mathcal{A}(x) : s := \sum_{i=1}^{d} \sum_{j=1}^{d} \sum_{k=1}^{d} \sum_{l=1}^{d} \mathcal{A}_{ijkl}(x) s_{ij} s_{kl}.$$

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- *(Cauchy) stress tensor*: $\sigma : H^1(\Omega;\mathbb{R}^d) \to L^2(\Omega;\mathbb{S}^d), \qquad u \mapsto \mathcal{A} : \varepsilon(u)$.

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$$\ell(v) := \int_\Omega f \cdot v \,\mathrm{d}x + \int_{\Gamma_N} F \cdot (\gamma v) \,\mathrm{d}s, \qquad v \in H^1(\Omega; \mathbb{R}^d)$$

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- *Internal forces* are described by the *stress* $\sigma(u) = \mathcal{A} : \varepsilon(u)$.
- *Work* $=$ *force* $\cdot$ *displacement*: The symmetric bilinear form

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- *Equilibrium* between both: $a(u, v) = \ell(v) \qquad \forall\, v \in H^1(\Omega; \mathbb{R}^d)$.

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- On $H^k(\Omega;\mathbb{R}^d)$, $k \in \mathbb{N}_0$, we denote the inner product and norm by

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- $H^0(\Omega;\mathbb{R}^d) = L^2(\Omega;\mathbb{R}^d)$.
- On $H^s(\Omega;\mathbb{R}^d)$ for $s = k + \theta$ with $k \in \mathbb{N}_0$ and $\theta \in (0,1)$ we use

$$(u,\, v)_{s,\Omega} := (u,\, v)_{k,\Omega} + \sum_{|\alpha|=k} \int_\Omega \int_\Omega \frac{\big(\mathrm{D}^\alpha u(x) - \mathrm{D}^\alpha u(y)\big) \cdot \big(\mathrm{D}^\alpha v(x) - \mathrm{D}^\alpha v(y)\big)}{|x-y|^{d+2\theta}} \,\mathrm{d}x\,\mathrm{d}y$$

and induce norm $\|\cdot\|_{s,\Omega}$.

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- We define the *classical trace operator* by

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and equip it with the norm

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- For $s = 1$ we have define $\ker(\gamma) =: H^1_0(\Omega; \mathbb{R}^d)$.

Admissible Displacements and the Nonpenetration Constraint

- *Homogeneous Dirichlet* boundary conditions on Γ_D are incorporated in the underlying Hilbert space

$$V := \left\{ v \in H^1(\Omega; \mathbb{R}^d) \;\middle|\; (\gamma v)|_{\Gamma_D} = 0 \right\} \subset H^1(\Omega; \mathbb{R}^d).$$

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- K is closed and convex in V.
- K is a *cone*, i.e. if $v \in K$ then $\lambda v \in K$ for all $\lambda \in [0,\infty)$.

Energy Minimization

- *Equilibrium*:

$$a(u,v) = \int_\Omega \sigma(u) : \varepsilon(v)\,\mathrm{d}x = \int_\Omega f \cdot v\,\mathrm{d}x + \int_{\Gamma_N} F \cdot (\gamma v)\,\mathrm{d}s, \qquad \forall\, v \in V$$

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Problem (Signorini's Problem – Energy Minimization)

Find $u \in K$ *such that:* $\qquad u \in \arg\min_{v \in K} \mathcal{J}(v)$

Standard Properties of a and ℓ

Proposition

The symmetric bilinear form $a : V \times V \to \mathbb{R}$ defined above is

i) bounded, *i.e. for all $v, w \in V$ there exists $C_1 < \infty$ such that*

$$|a(u,v)| \leq C_1 \|u\|_{1,\Omega} \|v\|_{1,\Omega} \qquad \textit{with} \qquad C_1 = C_{\mathcal{A}},$$

ii) V-elliptic, *i.e. for all $v \in V \setminus \{0\}$ there exist $C_2 > 0$ such that*

$$a(v,v) \geq C_2 \|v\|_{1,\Omega}^2 \qquad \textit{with} \qquad C_2 = \frac{\alpha_{\mathcal{A}}}{C_K^2}.$$

Moreover, the linear functional ℓ defined above is bounded, *i.e. for all $v \in V$ there exists $C_3 > 0$ such that*

$$|\ell(v)| \leq C_3 \|v\|_{1,\Omega} \qquad \textit{with} \qquad C_3 = \max\{1, C_{\mathrm{tr}}\} \left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right)$$

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Lemma ($\mathcal{J}$ is Coercive and w.l.s.c. on V)

i) $\mathcal{J}$ *is* coercive *on* V*, i.e.* $\mathcal{J}(v) \to \infty$ *as* $\|v\|_{1,\Omega} \to \infty$*.*

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Proof:

- By boundedness of ℓ we have $|\ell(v)| \leq \|\ell\|_{V'}\|v\|_{1,\Omega}$. Hence

$$\mathcal{J}(v) = \frac{1}{2}\,a(v,v) - \ell(v) \geq \frac{\alpha}{2}\,\|v\|_{1,\Omega}^2 - \|\ell\|_{V'}\,\|v\|_{1,\Omega} \xrightarrow{\|v\|_{1,\Omega}\to\infty} \infty.$$

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Lemma (W.l.s.c. of Norms)

Let $(X, \|\cdot\|_X)$ *be a normed space. Then* $\|\cdot\|_X : X \to \mathbb{R}_{\geq 0}$ *is a weakly lower semicontinuous functional, i.e. if* $x_j \rightharpoonup x$ *in* X, *then*

$$\|x\|_X \leq \liminf_{j\to\infty} \|x_j\|_X.$$

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Proof:

- On $L^2(\Omega; \mathbb{S}^d)$, we define the norm

$$\|s\|_{\mathcal{A}}^2 := \int_\Omega s : \mathcal{A} : s \,\mathrm{d}x, \qquad s \in L^2(\Omega; \mathbb{S}^d).$$

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- We have $a(v, v) = \|\varepsilon(v)\|_{\mathcal{A}}^2$.
- Since ε is linear and bounded, we have $\varepsilon(v_j) \rightharpoonup \varepsilon(v)$ in $L^2(\Omega; \mathbb{S}^d)$.

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- $\ell \in V'$ and thus $\ell(v_j) \to \ell(v)$.
- Therefore,

$$\mathcal{J}(v) = \frac{1}{2}a(v,v) - \ell(v) \leq \frac{1}{2}\liminf_{j\to\infty} a(v_j,v_j) - \lim_{j\to\infty} \ell(v_j) = \liminf_{j\to\infty} \mathcal{J}(v_j).$$

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Proposition (Existence of Minimizer of $\mathcal{J}$)

The functional $\mathcal{J}$ has at least one minimizer $u \in K$.

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- Since V is Hilbert space there exists subsequence $\{v_{j_k}\}_{k \in \mathbb{N}}$ and some $u \in V$ such that $v_{j_k} \rightharpoonup u$ in V.

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- $u \in K$ since K is (weakly) closed.
- By w.l.s.c. of $\mathcal{J}$, $u \in K$ is minimizer of $\mathcal{J}$ because
$$\mathcal{J}(u) \leq \liminf_{k \to \infty} \mathcal{J}(v_{j_k}) = M = \inf_{v \in K} \mathcal{J}(v).$$

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If there exists a minimizer of the functional $\mathcal{J}$, it is unique.

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If there exists a minimizer of the functional $\mathcal{J}$, it is unique.

Proof:

- $\mathcal{J}$ is strictly convex on V and thus on the convex set K. Thus, a minimizer $u \in K$ is unique. $\square$

A-Priori Stability Bound

Proposition (Stability estimate for the minimizer)

The Unique minimizer $u \in K$ of $\mathcal{J}$ satisfies the a-priori bound

$$\|u\|_{1,\Omega} \leq \frac{2\, C_K^2\, \max\{1, C_{\mathrm{tr}}\}}{\alpha_{\mathcal{A}}} \left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right).$$

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- Recall: a is V-elliptic with $a(u,u) \geq \alpha_{\mathcal{A}}/C_K^2\, \|u\|_{1,\Omega}^2$ and ℓ is bounded with

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- Recall: a is V-elliptic with $a(u,u) \geq \alpha_{\mathcal{A}}/C_K^2\,\|u\|_{1,\Omega}^2$ and ℓ is bounded with

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- Finally,

$$\frac{\alpha_{\mathcal{A}}}{2\,C_K^2}\,\|u\|_{1,\Omega}^2 \leq \frac{1}{2}\,a(u,u) \leq |\ell(u)| \leq \max\{1, C_{\mathrm{tr}}\}\,(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N})\,\|u\|_{1,\Omega}.$$

$\square$

Variational Inequality as an Optimality Condition

Proposition (First-order optimality condition on convex sets)

A function $u \in K$ minimizes $\mathcal{J}$ over K if and only if it satisfies the variational inequality

$$\mathcal{J}'(u)[v-u] = a(u, v-u) - \ell(v-u) \geq 0 \qquad \forall\, v \in K.$$

Proof: $\Longrightarrow$:

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- By chainrule, $\phi'(0^+) = \mathcal{J}'(u)[v-u] \geq 0$.

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Problem (Signorini's Problem – Weak Formulation 1)

Find $u \in K$ *such that:* $\qquad a(u, v-u) \geq \ell(v-u) \qquad \forall\, v \in K.$

Weak Formulations / Variational Inequality

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Find $u \in K$ *such that*

$$\begin{cases} a(u,u) = \ell(u), \\ a(u,v) \geq \ell(v) \qquad \forall\, v \in K. \end{cases}$$

Content

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- This assumptions is much stronger than needed!
- In strong formulations, we need an equilibrium equation $-\operatorname{div}\sigma(u) = f$ in $L^2(\Omega;\mathbb{R}^d)$.
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$$H(\operatorname{div};\Omega) := \left\{\tau \in L^2(\Omega;\mathbb{S}^d) \;\middle|\; \operatorname{div}\tau \in L^2(\Omega;\mathbb{R}^d)\right\},$$

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- Normal traces on $H(\mathrm{div};\Omega)$ are understood as follows.

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- In the smooth case, $\gamma_{\mathrm{n}}(\tau)$ coincides with the classical traction $\tau\,\mathrm{n}$. We thus write $\tau\,\mathrm{n} = \gamma_{\mathrm{n}}(\tau)$.

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- *Useful identity*:

$$a(u,v) - \ell(v) = \langle \sigma(u)\mathrm{n}, \gamma v\rangle_\Gamma - \int_{\Gamma_N} F \cdot (\gamma v)\;\mathrm{d}s, \qquad \forall\, v \in V.$$

A Hybrid Formulation

Problem (Signorini's Problem – Hybrid Formulation)

For given $f \in L^2(\Omega;\mathbb{R}^d)$ *and* $F \in L^2(\Gamma_N;\mathbb{R}^d)$, *find* $u \in K$ *such that* $\sigma(u) \in H(\mathrm{div};\Omega)$ *and*

$$\begin{cases} -\operatorname{div}\sigma(u) = f & \text{a.e. in } \Omega, & (1a)\\ \langle \sigma(u)\mathrm{n}, \gamma u\rangle_\Gamma = \displaystyle\int_{\Gamma_N} F\cdot(\gamma u)\,\mathrm{d}s, & & (1b)\\ \langle \sigma(u)\mathrm{n}, \gamma v\rangle_\Gamma \geq \displaystyle\int_{\Gamma_N} F\cdot(\gamma v)\,\mathrm{d}s & \forall\, v \in K. & (1c)\end{cases}$$

Proposition (Equivalence of Hybrid and Weak Formulation)

Assume $\sigma(u) \in H(\mathrm{div};\Omega)$. *Then* $u \in K$ *solves the hybrid problem above if and only if it solves the weak problem*

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Proof: *Hybrid* $\Rightarrow$ *Weak*:

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- Subtract (1b) from (1c):

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- Let $\varphi \in \mathcal{D}(\Omega;\mathbb{R}^d)$. Then $u \pm t\varphi \in K$ for $|t| \ll 1$, hence the weak inequality implies

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$$\text{Find } u \in K \text{ such that} \qquad a(u, v-u) \geq \ell(v-u) \qquad \forall v \in K.$$

Proof: *Weak $\Rightarrow$ Hybrid (Bulk Equation)*:

- Let $\varphi \in \mathcal{D}(\Omega;\mathbb{R}^d)$. Then $u \pm t\varphi \in K$ for $|t| \ll 1$, hence the weak inequality implies

$$a(u,\varphi) = \ell(\varphi) = \int_\Omega f \cdot \varphi \,\mathrm{d}x = \langle f, \varphi\rangle \qquad \forall\, \varphi \in \mathcal{D}(\Omega;\mathbb{R}^d).$$

- The left hand side is

$$a(u,\varphi) = \int_\Omega \sigma(u) : \varepsilon(\varphi) = \int_\Omega \sigma(u) : \nabla\varphi \,\mathrm{d}x = \langle -\operatorname{div}\sigma(u), \varphi\rangle.$$

- Thus, $-\operatorname{div}\sigma(u) = f$ in $\mathcal{D}'(\Omega;\mathbb{R}^d)$ and by $\sigma(u) \in H(\mathrm{div};\Omega)$, it holds in $L^2(\Omega;\mathbb{R}^d)$ as well.

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- This is exactly (1b).

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- This means

$$\langle \sigma(u)\mathrm{n}\,\gamma v\rangle_\Gamma \geq \int_{\Gamma_N} F\cdot\gamma v\,\mathrm{d}s + \underbrace{\left[\langle \sigma(u)\mathrm{n}\,\gamma u\rangle_\Gamma - \int_{\Gamma_N} F\cdot\gamma u\,\mathrm{d}s\right]}_{=0} \qquad \forall\, v \in V.$$

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$$\langle \gamma_{\mathrm{n},0}(\tau),\, g \rangle_{\Gamma_0} := \langle \gamma_{\mathrm{n}}(\tau),\, \mathrm{Ext}_0(g) \rangle_{\Gamma}, \qquad g \in H^{1/2}_{00}(\Gamma_0; \mathbb{R}^d),$$
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- Therefore the normal trace of $\tau \in H(\mathrm{div};\Omega)$ on Γ_C is the unique element $\gamma_n, C \in W_C'$ given by $\langle \gamma_{\mathrm{n},0}(\tau),\, g\rangle_{\Gamma_C}$

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- In general, $\gamma_{\mathrm{n},0}(\tau)$ need not extend continuously to all of W_C.

Classical Strong Formulation under Additional Regularity

Problem (Signorini's Problem – Classical Strong Formulation)

For given $f \in L^2(\Omega;\mathbb{R}^d)$ *and* $F \in L^2(\Gamma_N;\mathbb{R}^d)$ *find* $u \in H^s(\Omega;\mathbb{R}^d)$ *with* $s > 3/2$ *such that the* small strain elasticity equations

$$\begin{cases} -\operatorname{div}\sigma(u) = f & \text{in } \Omega, & (2a)\\ \sigma(u) = \mathcal{A} : \varepsilon(u) & \text{in } \Omega, & (2b)\\ u = 0 & \text{on } \Gamma_D, & (2c)\\ \sigma(u)\mathrm{n} = F & \text{on } \Gamma_N, & (2d)\end{cases}$$

and the contact conditions

$$\begin{cases} u_{\mathrm{n}} \leq 0 & \text{on } \Gamma_C, & (3a)\\ \sigma_{\mathrm{n}}(u) \leq 0 & \text{on } \Gamma_C, & (3b)\\ \sigma_{\mathrm{n}}(u)\, u_{\mathrm{n}} = 0 & \text{on } \Gamma_C, & (3c)\\ \sigma_t(u) = 0 & \text{on } \Gamma_C & (3d)\end{cases}$$

hold.